

Paper 2 - Set 08 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Functions and transformations
2	D	Quadratic functions
3	C	Simultaneous equations and inequalities
4	C	The logic of arguments
5	A	Identifying errors in proofs
6	A	Mathematical proof
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8	E	Quadratic functions
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10	A	Mathematical proof
11	B	The logic of arguments
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14	C	Exponentials and logarithms
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18	F	Trigonometric functions and equations
19	F	Functions and transformations
20	D	Chains of reasoning

Worked solutions

Question 1 FUN-08 · Functions and transformations · spec MM1.7

For which values of x is $\sqrt{x^2} = -x$?

Answer: C

$\sqrt{x^2} = |x|$, and $|x| = -x$ exactly when $x \leq 0$.

Question 2 QUA-10 · Quadratic functions · spec MM1.3

The quadratic $x^2 + px + q$ has minimum value -4 and one of its roots is 0. Which of the following could be the quadratic?

Answer: D

Minimum -4 means $x^2 + px + q = (x - h)^2 - 4$, with roots $h \pm 2$. One root is 0, so $h = \pm 2$: the quadratic is $x^2 - 4x$ or $x^2 + 4x$. Only $x^2 - 4x$ is offered.

Question 3 SIM-04 · Simultaneous equations and inequalities · spec MM1.4

The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .

Answer: C

$x^2 + (x + k)^2 = 8 \Rightarrow 2x^2 + 2kx + k^2 - 8 = 0$. Tangent when $4k^2 - 8(k^2 - 8) = 0 \Rightarrow -4k^2 + 64 = 0 \Rightarrow k = \pm 4$.

Question 4 LGC-08 · The logic of arguments · spec Arg4

Which of the following is the negation of '**there exists a positive integer n such that $n^2 + n + 41$ is not prime**'?

Answer: C

Negating 'there exists n with not $P(n)$ ' gives 'for all n , $P(n)$ '.

Question 5 ERR-04 · Identifying errors in proofs · spec Err1

Claim: for all real x , if $x^2 > 9$ then $x > 3$.

I Suppose $x^2 > 9$.

II Taking square roots of both sides, $x > 3$.

III This proves the claim.

Which of the following is true?

Answer: A

$x^2 > 9$ gives $|x| > 3$, i.e. $x > 3$ or $x < -3$. The claim itself is false, and line II is where the invalid deduction happens.

Question 6 PRF-03 · Mathematical proof · spec Prf1

Here is a proof that $\sqrt{2}$ is irrational, with its steps in the wrong order.

- I Then $p^2 = 2q^2$, so p^2 is even, so p is even; write $p = 2k$.
- II Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ where p and q are positive integers with no common factor.
- III This contradicts the assumption that p and q have no common factor, so $\sqrt{2}$ is irrational.
- IV Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.

Which is the correct order?

Answer: A

Assume (II), deduce p even (I), deduce q even (IV), reach the contradiction (III).

Question 7 GRA-05 · Graphs of functions · spec MM8.1

Which of the following statements about the graph of $y = |2x - 4|$ is/are true?

- I It meets the x -axis at exactly one point.
- II It is symmetric about the line $x = 2$.
- III It lies entirely on or above the line $y = 0$.

Answer: H

$|2x - 4| = 0$ only at $x = 2$ (I). $|2x - 4| = 2|x - 2|$ is symmetric about $x = 2$ (II). A modulus is never negative (III).

Question 8 X2-30 · Quadratic functions · spec MM1.3

The equation $x^2 + bx + c = 0$ has real coefficients. Which of the following is a **sufficient** condition for it to have two distinct real roots?

Answer: E

$c < 0$ makes the discriminant $b^2 - 4c$ positive. $b^2 \geq 4c$ allows equality (a repeated root); the others do not control the discriminant.

Question 9 LGC-02 · The logic of arguments · spec Arg2

Let P be the statement ' $x > 2$ ' and Q the statement ' $x^2 > 4$ ', where x is a real number. Which one of the following is true?

Answer: C

$x > 2 \Rightarrow x^2 > 4$, so P is sufficient for Q . But $x = -3$ gives $x^2 > 4$ without $x > 2$, so P is not necessary.

Question 10 PRF-16 · Mathematical proof · spec Prf4

A proof that **if a and b are positive real numbers then $\frac{a+b}{2} \geq \sqrt{ab}$ ** is given with its lines scrambled.

I Hence $a + b \geq 2\sqrt{ab}$, and dividing by 2 gives the result.

II Expanding, $a - 2\sqrt{ab} + b \geq 0$.

III Since squares are non-negative, $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

Which order gives a valid proof?

Answer: A

Start from a known truth (III), expand (II), rearrange (I). Starting from the result and working backwards (I, II, III) would need every step to be reversible and is presented in the wrong direction.

Question 11 LGC-23 · The logic of arguments · spec Arg1

Consider the statement: **if a and b are irrational, then $a + b$ is irrational.** Which of the following pairs is a counterexample?

Answer: B

A counterexample needs the hypothesis true (both irrational) and the conclusion false ($a + b$ rational):
 $\sqrt{2} + (-\sqrt{2}) = 0$.

Question 12 ERR-02 · Identifying errors in proofs · spec Err2

Here is a purported proof that $1 = 2$.

I Let $a = b$, with $a \neq 0$.

II Then $a^2 = ab$.

III So $a^2 - b^2 = ab - b^2$, i.e. $(a - b)(a + b) = b(a - b)$.

IV Dividing both sides by $(a - b)$: $a + b = b$.

V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

Answer: A

Since $a = b$, $a - b = 0$ and dividing by zero is invalid.

Question 13 PRF-08 · Mathematical proof · spec Prf2

It is given that for all real x : if $f(x) > 0$ then $g(x) > 0$. It is also known that $g(2) = -1$. What can be deduced?

Answer: D

Contrapositive: if $g(x) \leq 0$ then $f(x) \leq 0$. Since $g(2) = -1 \leq 0$, $f(2) \leq 0$ (it could be 0 or negative).

Question 14 LOG-07 · Exponentials and logarithms · spec MM5.2

Which of the following statements is/are true for all positive real x and y ?

- I $\log_{10}(x + y) = \log_{10} x + \log_{10} y$
- II $\log_{10}(xy)^2 = 2\log_{10} x + 2\log_{10} y$
- III $\log_{10} x - \log_{10} y = \log_{10} x \div \log_{10} y$

Answer: C

II is correct: $\log(xy)^2 = 2\log(xy) = 2\log x + 2\log y$. I and III confuse the laws: log of a sum has no simple form, and a difference of logs is log of a quotient, not a quotient of logs.

Question 15 CIR-11 · Circles and circle theorems · spec MM3.2

Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I The circles touch externally.
- II The circles intersect at exactly two points.
- III The point $(5, 0)$ lies on both circles.

Answer: F

Radii 5 and 3, centres 8 apart $= 5 + 3$, so the circles touch externally at one point (I true, II false). That point is $(5, 0)$: $25 = 25$ and $9 + 0 = 9$ (III true).

Question 16 NUM-12 · Number, units and estimation · spec M2.3

n is a positive integer such that n , $n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I There is exactly one such value of n .
- II n must be odd.
- III One of n , $n + 2$, $n + 4$ must be divisible by 3.

Answer: H

Among any three numbers of the form n , $n + 2$, $n + 4$ one is a multiple of 3 (they cover all residues mod 3) - III. So one of them must equal 3 itself for all to be prime, giving $n = 3$ only $(3, 5, 7)$ - I. And 3 is odd - II.

Question 17 RSN-17 · Chains of reasoning · spec Prf5

Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?

Answer: A

If Alice is truthful then Ben lies, so Cara is truthful: two truth-tellers, impossible. If Cara is truthful then Alice and Ben both lie; but Alice lying means Ben is truthful: contradiction. So Ben is the truth-teller: Cara lies (indeed Ben is not lying) and Alice lies (Ben is not lying), which is consistent.

Question 18 TRG-17 · Trigonometric functions and equations · spec MM4.5

Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

Answer: F

I is the Pythagorean identity. II is false (e.g. $x = \frac{\pi}{2}$; $0 \neq 2$). III: translating $\sin$ by half a period negates it, so III is true.

Question 19 FUN-10 · Functions and transformations · spec MM8.2

$f(x) = x^2 + 1$. Which of the following statements is/are true?

I $f(f(x)) = x^4 + 2x^2 + 2$ for all x

II $f(x + 1) = f(x) + 1$ for all x

III $f(2x) = 4f(x) - 3$ for all x

Answer: F

I: $f(f(x)) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$; true. II: $f(x + 1) = x^2 + 2x + 2 \neq x^2 + 2$; false. III: $f(2x) = 4x^2 + 1$ and $4f(x) - 3 = 4x^2 + 1$; true.

Question 20 RSN-01 · Chains of reasoning · spec Prf5

A selection of n distinct integers is taken from $1, 2, \dots, 20$. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

Answer: D

Pair up $\{1, 11\}, \{2, 12\}, \dots, \{10, 20\}$: ten pairs. Choosing 11 numbers forces two from the same pair (pigeonhole); choosing one from each pair gives 10 numbers with no such difference.